MINI PROJECT 2

NETWORK PROTOCOL ANALYSIS

OSI MODEL – THE SEVEN LAYERS OF NETWORK COMMUNICATION

Student Name	Vukoti Reethusri
Roll / Registration No.	Stu-36527D
Batch	Cyber Security/G17
Platform	Sure proED
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CYBER SECURITY PRACTICAL / MINI PROJECT

1. Introduction

The Open Systems Interconnection (OSI) model is a conceptual framework that explains how data travels between devices over a network. It divides network communication into seven layers, with each layer performing a specific function.

In this Cisco Packet Tracer practical, the OSI model is studied in relation to PCs, switches, routers, VLANs, IP addressing, and network communication. Understanding these layers helps in network troubleshooting and cybersecurity analysis.

2. Objectives

- To understand the seven layers of the OSI model.
- To identify the functions of each layer.
- To understand the roles of switches and routers.
- To understand IP addressing, MAC addressing, and port numbers.
- To study data encapsulation and decapsulation.
- To relate the OSI model to a Cisco Packet Tracer topology.
- To understand how the OSI model supports network troubleshooting and security.

3. The Seven Layers of the OSI Model

Layer 1: Physical Layer

The Physical Layer is the first and lowest layer of the OSI model. It transmits raw bits (0s and 1s) through a physical or wireless medium.

Functions:

- Transmits binary data through the communication medium.
- Defines cables, connectors, interfaces, and signal characteristics.
- Establishes physical connections between network devices.
- Supports electrical, optical, or radio signals.

Examples and devices: Examples: Ethernet cables, fibre-optic cables, and wireless signals. Devices: hubs, repeaters, cables, and physical network interfaces.

Application in the practical: Ethernet cables connect PCs to switches and network devices. Active links indicate that the physical connection is established.

Cybersecurity relevance: Unauthorized physical access can allow an attacker to connect a rogue device or tamper with network equipment.

Layer 2: Data Link Layer

The Data Link Layer provides communication between devices on the same local network segment.

Functions:

- Organizes data into frames.
- Uses MAC addresses for local network delivery.
- Performs error detection at the frame level.
- Supports switching and logical network segmentation through VLANs.

Examples and devices: Examples: Ethernet and Wi-Fi MAC. Devices: Ethernet switches and bridges.

Application in the practical: Switches forward Ethernet frames using MAC address tables. VLANs logically separate devices into different broadcast domains.

Cybersecurity relevance: MAC spoofing, MAC flooding, and VLAN-related attacks can target Layer 2.

Layer 3: Network Layer

The Network Layer handles logical addressing and the forwarding of packets between different IP networks.

Functions:

- Uses IP addresses to identify network interfaces.
- Determines routes to remote networks.
- Forwards packets between networks.
- Supports communication across different subnets.

Examples and devices: Protocols: IPv4, IPv6, and ICMP. Devices: routers and Layer 3 switches.

Application in the practical: Router interfaces use IP addresses and subnet masks. Static routes show how to reach remote networks. The ping command uses ICMP to test connectivity.

Cybersecurity relevance: IP spoofing, malicious routing changes, and network-layer denial-of-service attacks can affect communication.

Layer 4: Transport Layer

The Transport Layer provides communication between applications running on different devices.

Functions:

- Divides application data into transport-layer units.
- Uses port numbers to identify applications and services.
- Supports reliable, ordered delivery through TCP.
- Supports connectionless transport through UDP.

Examples and devices: Protocols: TCP and UDP. TCP is used by many web and file-transfer connections, while UDP is used by many DNS queries and real-time communication applications.

Application in the practical: Transport-layer communication can be examined by generating supported TCP or UDP traffic in Packet Tracer Simulation mode.

Cybersecurity relevance: Port scanning and attacks against exposed network services are commonly associated with the transport layer.

Layer 5: Session Layer

The Session Layer establishes, manages, and terminates communication sessions between applications.

Functions:

- Establishes communication sessions.
- Coordinates the exchange of information between applications.
- Maintains and manages ongoing sessions.
- Supports session recovery and synchronization where implemented.

Examples and devices: Examples: session management in remote communication and application systems.

Application in the practical: Session-layer concepts explain how applications maintain communication. In modern networks, these functions are often implemented within application protocols and software rather than as a separate device configuration.

Cybersecurity relevance: Session hijacking and session fixation can allow unauthorized use of a legitimate user's session.

Layer 6: Presentation Layer

The Presentation Layer deals with how data is represented and interpreted by applications.

Functions:

- Converts data between formats.
- Handles character encoding.
- Supports data serialization and representation.
- Provides encryption and compression functions in relevant implementations.

Examples and devices: Examples: UTF-8 text encoding, JPEG images, and structured data formats.

Application in the practical: This layer explains how data is represented before applications process it. Its functions are generally implemented by software and protocols, not configured as a separate Packet Tracer network device.

Cybersecurity relevance: Encryption and correct data handling help protect sensitive information from unauthorized access and interpretation.

Layer 7: Application Layer

The Application Layer is the topmost layer of the OSI model. It provides network services to applications used by people and organizations.

Functions:

- Supports web access and file transfer.
- Enables domain name resolution.
- Supports email and other network services.
- Defines application-level requests and responses.

Examples and devices: Protocols: HTTP, HTTPS, DNS, SMTP, and FTP.

Application in the practical: Supported application services, such as HTTP or DNS, can be tested in Packet Tracer if the relevant services are configured.

Cybersecurity relevance: Phishing, malicious web requests, insecure application configurations, and application vulnerabilities can target this layer.

4. Summary of the Seven Layers

Layer	Name	Data Unit	Main Concept
7	Application	Data	Network services
6	Presentation	Data	Data representation
5	Session	Data	Session management
4	Transport	Segment / Datagram	TCP, UDP, and ports
3	Network	Packet	IP addressing and routing
2	Data Link	Frame	MAC addressing and switching
1	Physical	Bits	Cables and signals

5. Difference Between OSI and TCP/IP Model

Feature	OSI Model	TCP/IP Model
Full form	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Number of layers	7 layers	4 layers
Developed by	ISO	DARPA
Nature	Reference model	Practical communication model
Usage	Understanding and designing networks	Real-world internet communication
Protocol dependency	Protocol-independent	Based on TCP/IP protocols
Example	Helps explain how network communication works	Used when browsing websites or sending emails

6. Basic Network Concepts — Simple Explanation

A network means connecting devices so they can share information. For example, a mobile connects to Wi-Fi and opens a website or video service.

Concept	Simple meaning	Example
Network	Connected devices	College computer lab
Node	A device in a network	Computer, printer
Client	Requests information	Your mobile
Server	Provides information	Website server
IP Address	Logical address of a device	192.168.1.10
MAC Address	Hardware network address	Address of a Wi-Fi adapter
Protocol	Rules for communication	TCP, HTTP
Packet	A small unit of data	Part of a message
Bandwidth	Data transferred per second	100 Mbps
Latency	Time taken for data to travel	20 milliseconds
DNS	Converts website names into IP addresses	google.com
DHCP	Automatically gives IP addresses	Connecting to Wi-Fi
Port	Identifies an application or service	HTTPS uses port 443
Gateway	Provides a path to another network	Your router
Firewall	Controls allowed and blocked traffic	Blocks unwanted connections

Types of Networks

- PAN: Connects personal devices, such as a phone and smartwatch.
- LAN: Connects devices in a home, office, or college.
- MAN: Connects networks across a city.
- WAN: Connects networks across large distances, such as the Internet.

Network Devices

Device	What it does
Router	Connects different networks.
Switch	Connects devices in a LAN.
Hub	Sends incoming data to all other connected ports.
Modem / ONT	Connects a network to the Internet Service Provider (ISP).
Access Point	Allows devices to connect through Wi-Fi.
Firewall	Protects by filtering traffic.
Repeater	Extends a signal.
Bridge	Connects LAN segments.
Gateway	Connects networks and may translate protocols.
NIC	Allows a device to connect to a network.

7. How Data Travels Through a Network

Imagine sending a message from a computer to a website:

- The computer creates and sends the data.
- A switch forwards frames inside the local area network (LAN).
- A router forwards packets toward another network.
- The website server receives and processes the request.
- The response returns through the network to the requesting device.

Types of Communication

- Unicast: One device sends to one device.
- Broadcast: One device sends to all devices in the local broadcast network.
- Multicast: One device sends to a selected group.
- Anycast: Data goes to one suitable device among several sharing the same destination address.

Important Protocols

Protocol	Simple meaning
HTTP / HTTPS	Used to access websites.
TCP	Delivers data reliably and in order.
UDP	Sends data with less overhead.
IP	Helps data reach the correct network.
ARP	Finds the MAC address for a local IPv4 destination.
DNS	Finds the IP address of a website.
DHCP	Automatically assigns IP settings.
ICMP	Used for diagnostics, such as ping.

8. Types of Network Devices and Their Communication

Router — Routing: Sends data between different networks.

Switch — Switching: Sends data to the correct device in a LAN using MAC addresses.

Hub — Broadcasting: Sends incoming data to all other connected ports.

Modem / ONT — Connectivity: Connects your network to the Internet Service Provider (ISP).

Access Point — Wireless: Allows devices to connect to a network through Wi-Fi.

Firewall — Security: Allows or blocks network traffic based on security rules.

Repeater — Regeneration: Regenerates a signal to extend its reach.

Bridge — Bridging: Connects LAN segments at the data-link layer.

Gateway — Translation: Can translate between different protocols; a default gateway also provides a route to other networks.

NIC — Networking: Allows a computer to send and receive network data.

Hub

A hub connects multiple computers in the same LAN. When it receives data from one computer, it sends that data to all other connected ports rather than forwarding it only to the intended device.

Switch

A switch primarily operates at Layer 2 of the OSI model. It connects devices within a local area network and forwards Ethernet frames using MAC address information. In this practical, switches connect PCs and support VLAN-based network segmentation.

Router

A router operates at Layer 3 of the OSI model. It connects different IP networks and forwards packets according to destination IP addresses and routing information. In this practical, routers connect different networks, provide gateway addresses, and use static routes to reach remote subnets.

Key difference: A switch primarily forwards Ethernet frames within a LAN, whereas a router forwards IP packets between networks.

NAT (Network Address Translation)

NAT allows a router to translate private IP addresses into a different IP address, often a public IP address, so devices can communicate with external networks.

9. Practical Example: Switch, Router, and NAT in Cisco Packet Tracer

The reference document describes a Packet Tracer exercise involving end devices, switches, routers, server services, IP configuration, serial links, static routing, and web-browser testing. The following steps summarize that workflow without claiming that a new Packet Tracer topology was executed.

- Download and install Cisco Packet Tracer from the Cisco Networking Academy resource page.
- Open the PC configuration window and select IP Configuration.
- On the server, open Services and enable HTTP and HTTPS if those services are part of the exercise; edit the index page if needed.
- Configure end devices with IP addresses, subnet masks, and the appropriate default gateway for their subnet.
- Configure router interfaces with the matching IP addresses and enable the interfaces.
- Configure the serial link between routers with the addresses assigned for the inter-router connection, then enable the interfaces.
- Add static routes so each router can reach the remote subnet through the next-hop router.
- From a PC command prompt, use ping to test connectivity to the remote host or server.

- Open the PC web browser and test the configured server using its IP address or hostname, as appropriate.

Important: Use the actual IP addresses and subnet masks assigned in your own Packet Tracer topology. A default gateway must be a valid router-interface address in the same subnet; it is not the network address itself.

10. Encapsulation and Decapsulation

When data is sent, each OSI layer provides information needed by the next stage of communication. This process is commonly described as encapsulation. At the receiving end, the information is interpreted and removed layer by layer; this is decapsulation.

OSI layer (sending side)	Typical data unit	What is added or handled
Application / Presentation / Session	Data	Application information and representation
Transport	Segment (TCP) / Datagram (UDP)	Transport header, including port information
Network	Packet	IP header with source and destination IP addresses
Data Link	Frame	Frame header/trailer, including MAC addressing where applicable
Physical	Bits	Transmission as electrical, optical, or radio signals

11. Troubleshooting Using the OSI Model

Layer	Example check
1 — Physical	Check power, cables, wireless signal, link lights, and interface status.
2 — Data Link	Check switch ports, VLAN membership, MAC learning, and local connectivity.
3 — Network	Check IP address, subnet mask, default gateway, routes, and ping results.
4 — Transport	Check TCP/UDP ports, service reachability, and filtering rules.
5 — Session	Check whether the application session is established and remains active.
6 — Presentation	Check data format, encoding, encryption, and certificate-related issues where relevant.
7 — Application	Check DNS, web/server service status, URLs, application settings, and error messages.

12. Conclusion

The OSI model provides a structured way to understand communication between network devices. Each of the seven layers has a distinct role, from transmitting bits at the Physical Layer to providing network services at the Application Layer. Relating the layers to switches, routers, addressing, ports, and protocols helps make network troubleshooting more systematic.

This mini project summarizes how OSI concepts can be applied to a Cisco Packet Tracer exercise and highlights security concerns at different layers. Successful practical results should be confirmed by testing the actual topology and recording the observed output.

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